STANDARDS-COMPLIANT OPTICAL FIBER LINK DESIGN

From Communication Requirement to Deployment Decision

ECA1804 – Optoelectronics and Optical Communication
CO1 Engineering Design Assignment

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Abstract

This report develops an engineering design for a 60 km optical-fiber link operating at 1550 nm and 10 Gb/s, with a possible future upgrade to 100 Gb/s. Two candidate fibers are evaluated using numerical aperture, V-number and modal behaviour. The design then examines the optical power budget, chromatic dispersion, polarization-mode dispersion (PMD), birefringence and optical signal-to-noise ratio (OSNR). Calculations are used to select a practical fiber arrangement and to identify changes needed before deployment. The final decision considers both the current 10 Gb/s service and the

Given Design Requirements

Parameter	Given value
Link distance	60 km
Operating wavelength	1550 nm
Data rate	10 Gb/s; provision for 100 Gb/s
Fiber A	Step-index; $n_1 = 1.4500$, $n_2 = 1.4450$, core radius a = 4.0 μm
Fiber B	Step-index; $n_1 = 1.480$, $n_2 = 1.465$, core radius a = 25 μm
Fiber attenuation components	Absorption = 0.04 dB/km; Rayleigh scattering = 0.11 dB/km; distributed bending = 0.03 dB/km
Transmitter power	+3 dBm
Connectors / splices	4 connectors × 0.4 dB; 10 splices × 0.08 dB
Building-entry bend loss	G.652.D: 0.8 dB; G.657.A: 0.1 dB
Receiver sensitivity / margin	−20 dBm / 6 dB
Chromatic dispersion	G.652.D: 17 ps/(nm·km); G.655: 4.5 ps/(nm·km); $\Delta\lambda = 0.4$ nm

Input pulse / bit period	30 ps / 100 ps at 10 Gb/s
DCF	D_DCF = −100 ps/(nm·km); attenuation = 0.5 dB/km
PMD coefficient / limit	0.1 ps/√km; limit = 0.1 T_b
OSNR data	Signal = −16 dBm; ASE = −39 dBm; required OSNR = 17 dB
Amplifier	Signal gain = 14 dB; ASE-noise increase = 16 dB

Stage 1 – Interpret the Communication Requirement

The link must carry 10 Gb/s over 60 km at 1550 nm while maintaining received power, pulse integrity, polarization performance and signal quality. The design should also permit an upgrade to 100 Gb/s. A single-mode fiber is the suitable propagation class for this distance and rate because it avoids intermodal dispersion. A bend-insensitive G.657.A fiber is attractive at the building entry, while G.652.D is a common standard single-mode transmission option. G.655 can reduce chromatic dispersion around 1550 nm.

Why 1550 nm?

The 1550 nm region is widely used in optical communications because silica fiber has relatively low attenuation there and optical amplifiers such as erbium-doped fiber amplifiers can operate in this wavelength band. Light behaves as an electromagnetic wave in the fiber; ray optics can give an intuitive picture, but wave optics and guided-mode analysis are needed to predict single-mode operation, cutoff and polarization effects. Coherence affects interference-sensitive systems, while polarization evolves due to fiber birefringence.

Design Requirement Statement

Design a standards-compliant 60 km single-mode optical link at 1550 nm that supports 10 Gb/s, maintains at least 6 dB receiver margin, limits dispersion and PMD, tolerates tight entry bends, meets the required OSNR of 17 dB, and provides a credible upgrade path to 100 Gb/s.

Stage 2 – Fiber and Propagation Configuration

Numerical aperture (NA)

For a step-index fiber in air, $NA = \sqrt{(n_1^2 - n_2^2)}$.

$$\textbf{Fiber A: } NA = \sqrt{(1.4500^2 - 1.4450^2)} = 0.1203$$

$$\textbf{Fiber B: } NA = \sqrt{(1.480^2 - 1.465^2)} = 0.2102$$

Fiber B accepts light over a larger cone because its NA is larger. NA describes acceptance, but a large NA does not by itself make a fiber better for long-distance high-speed transmission.

V-number and mode behaviour

The normalized frequency is $V = (2\pi a/\lambda)\, NA$. At $\lambda = 1.55\ \mu m$:

$$\textbf{Fiber A: } V = (2\pi \times 4.0 / 1.55) \times 0.1203 \approx 1.95$$

$$\textbf{Fiber B: } V = (2\pi \times 25 / 1.55) \times 0.2102 \approx 21.26$$

A step-index fiber is single-mode when $V < 2.405$. Therefore, Fiber A is single-mode at 1550 nm. Fiber B is strongly multimode. For a step-index multimode fiber, the approximate mode count is $M \approx V^2/2$, giving $M \approx$

$21.26^2/2 \approx 226$ modes (approximate). For a graded-index multimode fiber, a common approximation is $M \approx V^2/4$, but Fiber B is specified as step-index.

For Fiber A, the single-mode cutoff occurs at $V = 2.405$. Since V is inversely proportional to wavelength, $\lambda_c = 2\pi a \cdot NA/2.405 \approx 1.26$ μm. At 1550 nm, $\lambda > \lambda_c$, so the fiber is single-mode. Ray theory is useful for qualitative acceptance and reflection ideas, but wave optics / LP-mode analysis is required for the modal cutoff decision.

Fiber-category comparison and decision

Option	Advantages	Limitations / use
G.652.D single-mode	Widely used standard single-mode fiber; compatible with 1550 nm systems	Specified building-entry bend loss is higher (0.8 dB)
G.655 single-mode	Lower dispersion in the given data: 4.5 ps/(nm·km)	Must check actual standard/vendor compatibility and system parameters
G.657.A bend-insensitive single-mode	Best of given options for tight bends; entry bend loss only 0.1 dB	Confirm compatibility and attenuation/dispersion specifications for selected product
Step-index multimode Fiber B	Large NA and core	About 226 modes; modal dispersion makes it unsuitable for this 60 km high-speed link

Design Decision 1: I recommend single-mode propagation using a G.657.A bend-insensitive fiber for the building entry, with a compatible standard single-mode transmission fiber selected for the main route. If one category must be named for the complete route from the supplied choices, G.655 is attractive for its lower specified chromatic dispersion, but its bend tolerance is not specified in the assignment. A practical deployment may use G.655 or G.652.D on the route and G.657.A for bend-sensitive sections, subject to compatibility checks.

Stage 3 – Optical Power Budget

Fiber attenuation

$$\alpha_{total} = \alpha_{absorption} + \alpha_{scattering} + \alpha_{bending}$$

$$\alpha_{total} = 0.04 + 0.11 + 0.03 = 0.18 \text{ dB/km}$$

$$60 \text{ km fiber loss} = 0.18 \times 60 = 10.8 \text{ dB}$$

The largest specified distributed mechanism is Rayleigh scattering (0.11 dB/km), contributing 6.6 dB over 60 km. Absorption contributes 2.4 dB and distributed bending contributes 1.8 dB.

Connector and splice losses

$$\text{Connector loss} = 4 \times 0.4 = 1.6 \text{ dB}$$

$$\text{Splice loss} = 10 \times 0.08 = 0.8 \text{ dB}$$

$$\text{Fixed connection loss} = 1.6 + 0.8 = 2.4 \text{ dB}$$

Budget item	G.652.D entry	G.657.A entry

Transmitter power	+3 dBm	+3 dBm
60 km fiber loss	10.8 dB	10.8 dB
Connector loss	1.6 dB	1.6 dB
Splice loss	0.8 dB	0.8 dB
Building-entry bend loss	0.8 dB	0.1 dB
Total link loss	14.2 dB	13.5 dB
Received power	−11.2 dBm	−10.5 dBm
Margin above −20 dBm sensitivity	8.8 dB	9.5 dB
Margin remaining after required 6 dB	2.8 dB	3.5 dB
Meets 6 dB requirement?	Yes, on supplied budget	Yes, on supplied budget

Received power is P_rx = P_tx − total loss. The available margin is P_rx − receiver sensitivity. The margin beyond the required system margin is available margin − 6 dB. Both budgets meet the stated 6 dB requirement under the given assumptions, and G.657.A saves 0.7 dB compared with G.652.D at the building entry.

Practical loss-reduction actions: (1) use G.657.A or maintain the manufacturer's minimum bend radius at entry points; (2) clean and inspect connectors and improve splice workmanship. Additional actions include minimizing unnecessary connection points and measuring insertion loss during commissioning.

Design Decision 2: I recommend the G.657.A bend-insensitive entry configuration because it reduces entry loss from 0.8 dB to 0.1 dB and increases the calculated system margin to 9.5 dB. The dominant specified distributed loss is Rayleigh scattering. The budget is acceptable against the stated 6 dB requirement, subject to real component specifications and commissioning measurements.

Stage 4 – Dispersion and PMD Management

Chromatic dispersion and pulse broadening

Chromatic dispersion broadening is $\Delta\tau = |D|\,\Delta\lambda\,L$.

$$\textbf{G.652.D: } \Delta\tau = 17 \times 0.4 \times 60 = 408 \text{ ps}$$

$$\textbf{G.655: } \Delta\tau = 4.5 \times 0.4 \times 60 = 108 \text{ ps}$$

Combining the 30 ps input pulse with broadening using root-sum-square (RSS), $\tau_{total} = \sqrt{(\tau_{in}^2 + \Delta\tau^2)}$:

$$\textbf{G.652.D: } \tau_{total} = \sqrt{(30^2 + 408^2)} \approx 409.1 \text{ ps}$$

$$\textbf{G.655: } \tau_{total} = \sqrt{(30^2 + 108^2)} \approx 112.1 \text{ ps}$$

The 10 Gb/s bit period is 100 ps. Both estimates exceed one bit period, especially G.652.D, indicating substantial pulse spreading and potential inter-symbol interference. G.655 is better in the given comparison, but 112.1 ps is still slightly greater than the bit period; the real transmitter spectral shape, receiver, modulation format and dispersion tolerance must be checked. This simple RSS method is the assignment's requested comparison, not a complete waveform simulation.

Dispersion compensation fiber (DCF)

First-order compensation requires the total accumulated dispersion to be approximately zero: D_main L_main + D_DCF L_DCF = 0.

For G.652.D: $L_{DCF} = -(17 \times 60)/(-100) = 10.2$ km

Added DCF loss = $10.2 \times 0.5 = 5.1$ dB

Using the G.657.A entry budget as the base (13.5 dB), adding the DCF loss gives total loss = 18.6 dB and received power = +3 − 18.6 = −15.6 dBm. Margin above sensitivity = −15.6 − (−20) = 4.4 dB, which is 1.6 dB below the required 6 dB system margin. Using the G.652.D entry budget instead gives total loss 19.3 dB, received power −16.3 dBm and sensitivity margin 3.7 dB.

Therefore, conventional DCF can correct first-order dispersion but its specified 5.1 dB loss makes the supplied power budget fail the 6 dB margin target unless the link is redesigned (e.g., lower-loss compensation, amplifier placement with OSNR verification, or a lower-dispersion transmission fiber).

Polarization-mode dispersion and birefringence

$PMD = D_{PMD}\sqrt{L} = 0.1 \times \sqrt{60} \approx 0.775$ ps

PMD limit at 10 Gb/s = 0.1×100 ps = 10 ps → passes

PMD limit at 100 Gb/s ($T_b \approx 10$ ps) = 0.1×10 ps = 1 ps → passes nominally

The calculated PMD is below both assignment limits, but the 100 Gb/s margin is small (about 0.225 ps), and a real upgrade requires vendor PMD specifications and system-level validation. A mitigation technique is to use low-PMD fiber and transceivers with electronic/digital compensation where supported.

Birefringence means the effective refractive index differs for two orthogonal polarization states. Beat length is $L_B = \lambda/B$. The assignment's exponent formatting for B is unclear in the extracted table; interpreting the intended values as $B = 1 \times 10^{-7}$ for standard fiber and $B = 4 \times 10^{-4}$ for PM fiber:

Standard fiber: $L_B = 1.55 \times 10^{-6} / 10^{-7} = 15.5$ m

PM fiber: $L_B = 1.55 \times 10^{-6} / (4 \times 10^{-4}) = 3.875$ mm

These beat-length values depend strongly on the assumed exponents, which are not clearly legible in the source PDF. Polarization-maintaining (PM) fiber has deliberately high birefringence and is typically used where a stable polarization state is required, such as in specialized optical sources/sensors. It is not normally required throughout a standard telecom link; low-PMD transmission fiber is the more relevant requirement.

Design Decision 3: I recommend a low-dispersion single-mode route (G.655 according to the supplied D value) and bend-insensitive G.657.A at the building entry. If G.652.D is selected for the main route, the calculated DCF length is 10.2 km, but its 5.1 dB loss causes the power margin to fail. Therefore, do not add this DCF without redesigning the power/OSNR budget. PMD is acceptable at 10 Gb/s and nominally below the given 100 Gb/s limit, but upgrade validation is mandatory.

Stage 5 – OSNR and Amplification Strategy

OSNR is the optical signal-to-noise ratio, normally expressed in dB and referenced here to a 0.1 nm optical bandwidth.

Initial OSNR = $P_{signal} - P_{ASE} = (-16) - (-39) = 23$ dB

After the amplifier, signal power becomes −16 + 14 = −2 dBm and ASE noise becomes −39 + 16 = −23 dBm.

Post-amplifier OSNR = (−2) − (−23) = 21 dB

OSNR change = 21 − 23 = −2 dB; margin over 17 dB requirement = 4 dB

The amplifier increases signal power by 14 dB but increases ASE noise by 16 dB. Consequently, OSNR degrades by 2 dB. The resulting 21 dB remains 4 dB above the specified 17 dB requirement, so it meets the

numerical requirement for this stage. However, OSNR should be verified at the actual receiver after all losses and amplification stages; OSNR alone does not guarantee BER because modulation format, filtering, receiver noise and nonlinear effects also matter.

Inadequate OSNR reduces signal-to-noise quality and Q-factor and generally increases BER, reducing system margin. Practical improvement measures include (1) reducing passive loss and unnecessary amplifier stages; (2) selecting a low-noise amplifier and optimizing gain and placement. For higher-rate coherent systems, suitable modulation, receiver DSP and adequate launch-power planning are also important.

Design Decision 4: I recommend accepting the stated amplifier only as a provisional strategy because the calculated post-amplifier OSNR is 21 dB, above the 17 dB requirement by 4 dB. It does not improve OSNR; it reduces it by 2 dB. Confirm OSNR at the receiver with the complete link and the intended 100 Gb/s equipment before deployment.

Stage 6 – Complete Link Verification

Requirement	Target	Calculated / selected result	Satisfactory?
Modal condition / fiber category	Single-mode for 60 km high speed	Fiber A V ≈ 1.95; single-mode; choose single-mode ITU-T route, G.657.A at bends	Yes
Distributed fiber loss	Given component losses	0.18 dB/km; 10.8 dB over 60 km	As specified
Power and margin	At least 6 dB margin	G.657.A entry case: P_rx = −10.5 dBm; margin = 9.5 dB	Yes without DCF
Pulse broadening	Compatible with 100 ps bit period	G.652.D RSS ≈ 409.1 ps; G.655 RSS ≈ 112.1 ps	No, without dispersion management / system validation
DCF power penalty	Maintain 6 dB margin if used	10.2 km DCF adds 5.1 dB; G.657 entry margin falls to 4.4 dB	No, needs redesign
PMD	≤ 10 ps at 10 Gb/s; ≤ 1 ps at 100 Gb/s	0.775 ps	Yes nominally; verify upgrade
OSNR	At least 17 dB	23 dB initially; 21 dB after amplifier	Yes for stated OSNR case

Final Parameter / Design Table

Design parameter	Adopted value / decision	Reason
Wavelength	1550 nm	Given operating condition; suitable low-loss telecom window
Main propagation	Single-mode fiber	Avoids intermodal dispersion and supports long distance/high bit rate
Bend-sensitive section	G.657.A-type fiber	Building-entry bend loss is 0.1 dB versus 0.8 dB for G.652.D
Main-route category	Prefer low-dispersion single-mode (G.655 per given D), subject to product compatibility	Given dispersion is 4.5 ps/(nm·km)
Total distributed attenuation	0.18 dB/km; 10.8 dB over 60 km	Sum of supplied absorption, scattering and bending coefficients
Connection loss	2.4 dB	Four connectors plus ten splices

Entry loss	0.1 dB	G.657.A option
Base received power	−10.5 dBm	Without DCF or amplifier, based on supplied budget
DCF length	Not included in final baseline; if G.652.D compensation is used, calculated 10.2 km	Specified DCF loss breaks the 6 dB margin without redesign
Amplifier	14 dB signal gain, +16 dB ASE increase (given)	Post-amplifier OSNR = 21 dB, 4 dB above requirement
PMD	Calculated 0.775 ps	Below stated 10 Gb/s and nominal 100 Gb/s limits

Proposed Link Block Diagram

10 Gb/s Transmitter
(+3 dBm, 1550 nm)
↓
Launch / Connector
↓
Single-mode transmission fiber (60 km)
(G.655 or compatible low-dispersion category)
↓
Splices / connector interfaces
↓
G.657.A bend-insensitive building-entry section
↓
Optical preamplifier / gain stage (only if validated)
↓
Optical receiver (sensitivity −20 dBm)
↓
BER / OSNR / dispersion verification

Note: the block diagram is conceptual. Exact amplifier location, span segmentation, optical filtering, transmitter launch conditions, receiver type and connector placement must be finalized from equipment datasheets and a detailed link engineering study. The given 60 km distance and component counts are used in the calculations above.

Final Engineering Decision – Would You Deploy This Design?

NO — not in the present form without further engineering verification. The basic single-mode power budget meets the 6 dB margin when the G.657.A entry loss is used, and the supplied amplifier OSNR calculation remains above the required 17 dB. However, the uncompensated pulse-broadening estimates exceed the 100 ps bit period for both given dispersion values. The calculated G.652.D DCF section is 10.2 km and adds 5.1 dB loss, reducing the power margin below the required 6 dB. Thus, the design cannot be approved for deployment solely on the basis of the baseline power budget.

Recommended redesign:

- Use single-mode fiber for the 60 km route, with G.657.A in bend-sensitive entry sections and a transmission fiber chosen after confirming actual attenuation, dispersion, PMD and compatibility specifications.
- Prefer the lower-dispersion option in the supplied data (G.655) if its actual product specifications and network compatibility are confirmed; evaluate whether electronic dispersion compensation or a suitable lower-loss compensation approach is preferable to the specified DCF.
- If DCF is required, include its 5.1 dB loss in the end-to-end power budget and redesign gain placement / passive losses so at least 6 dB receiver margin remains. Verify the corresponding OSNR after amplification.
- Optimize connectors, splices and bend routing; use inspection and optical loss testing during installation.
- For a 100 Gb/s upgrade, validate the selected transceivers, modulation format, chromatic dispersion tolerance, PMD, OSNR and BER with vendor specifications and link simulation/measurement.

The design may be deployed only after dispersion management and the combined power/OSNR budget are re-engineered and verified. The final acceptance tests should include end-to-end insertion loss, received power, OSNR in 0.1 nm bandwidth, dispersion/PMD tolerance and BER.

Conclusion

The calculations show that fiber A is single-mode at 1550 nm ($V \approx 1.95$), while fiber B is multimode ($V \approx 21.26$, approximately 226 guided modes under the step-index approximation). The combined specified fiber attenuation is 0.18 dB/km, producing 10.8 dB loss over 60 km. With G.657.A entry loss, the baseline received power is −10.5 dBm and the sensitivity margin is 9.5 dB. Chromatic dispersion is the key unresolved pulse-integrity issue: estimated RSS pulse widths are approximately 409.1 ps for G.652.D and 112.1 ps for G.655, compared with a 100 ps bit period. The G.652.D DCF length of 10.2 km adds 5.1 dB and makes the supplied 6 dB power-margin requirement fail unless the design is changed. The stated amplifier yields 21 dB OSNR, meeting the 17 dB target but worsening OSNR by 2 dB. Therefore, the technically justified decision is not to deploy until dispersion management, power margin and OSNR are jointly validated.

References / Basis

Primary source: ECA18 – Optoelectronics and Optical Communication, CO1 Engineering Design Assignment PDF supplied with the task. All numerical inputs in this report are taken from that assignment. Standard formulas used: $NA = \sqrt{(n_1^2 - n_2^2)}$; $V = (2\pi a/\lambda)NA$; step-index multimode mode count $M \approx V^2/2$; α_total = sum of attenuation coefficients; $\Delta\tau = |D|\Delta\lambda L$; RSS pulse width = $\sqrt{(\tau_in^2 + \Delta\tau^2)}$; DCF compensation condition $D_1L_1 + D_DCF\ L_DCF = 0$; $PMD = D_PMD\sqrt{L}$; $OSNR(dB) = P_signal(dBm) - P_ASE(dBm)$.